

RES-5 sunset-norm certificate, v1.1 (audit re-issue): the self-energy/

TECT verification-first repository · claim B1-RH-ENUM

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1. Purpose and scope

This is an audit re-issue of v1.0 following the operator adversarial review of 2026-06-09. The review accepts the self-energy/free-energy normalization correction (Section 2) and retracts the survival verdict (Sections 4–5): the higher-skeleton tail bound carried by v1.0 is screened-finite but *same order* as the leading skeleton, and is therefore not bounded into the thin SC-SCOPE slack. The honest status of RES-5/GAP-2 is OPEN. No tier action; B1 stays T6 CONDITIONAL on {H-LAYER}.

2. Correction (ACCEPTED): the certificate quantity is a free-energy ratio

The a0-skeleton estimate $ca_0 \sim 0.49 \cdot a_0 \cdot (\|\Sigma_2^{\text{pd}}\|/\Delta F_{\text{margin}})$ took $\|\Sigma_2^{\text{pd}}\|/\Delta F_{\text{margin}} \sim 0.04$, but that 0.04 is the FREE-ENERGY (sunset) ratio, not a self-energy norm. Because $\Delta\Gamma_2^{\text{pd}} = \|\Sigma_2^{\text{pd}}\| \|\delta G_*^{\text{pd}}\|$, quoting the free-energy ratio AND multiplying by $\|\delta G_*^{\text{pd}}\| = O(a_0)$ double-counts a_0 . The certificate quantity is directly

$$\mathcal{R}_{\text{cert}} = \frac{|\Delta\Gamma_2^{\text{pd}}|}{\Delta F_{\text{margin}}},$$

whose leading (sunset / 2-loop) value is the SC-SCOPE certified joint, NOT 0.2%. This correction stands and is the accepted content of this note.

3. The normalization map (leading skeleton; identity still OPEN)

SC-SCOPE certified the endpoint joint

$$\text{joint} = \frac{\text{MARGIN}}{C_2 + C_{\text{sunset}} + C_{\text{quartic}}} = \times 1.040 \text{ (certified)} \longrightarrow \times 1.13 \text{ (sunset cap)},$$

both > 1 : the third cumulant (the leading 2PI skeleton) does not overturn the one-loop selection. Structurally this is the leading-skeleton statement $\mathcal{R}_{\text{cert}}^{\text{leading}} < 1$. The exact identity $\mathcal{R}_{\text{cert}}^{\text{leading}} = \mathcal{C}_{\text{SC}}$, with $\mathcal{C}_{\text{SC}}$ the SC-SCOPE $C_{\text{sunset}} + C_{\text{quartic}}$ normalization, is asserted from the shared third-cumulant origin but is NOT proved here. It is the first residual.

4. Higher skeletons: screened-finite but SAME ORDER (corrected reading)

v1.0 called the $\ell \geq 3$ tail “screened sub-dominant”. The 2PI screened response (res5-a0-skeleton-sensitivity) bounds each order by the previous times $1/(1+g) = 0.49$, giving the geometric tail

$$C_{\text{higher}} = \sum_{\ell \geq 3} |\Delta\Gamma_{\ell}^{\text{pd}}| \lesssim |\Delta\Gamma_{2\text{-loop}}^{\text{pd}}| \frac{1}{1-0.49} \approx 2 |\Delta\Gamma_{2\text{-loop}}^{\text{pd}}|.$$

This is *screened-finite*, but it is the SAME ORDER as the leading skeleton, not a small fraction of it. The selection slack at the thin joint is

$$\frac{\text{MARGIN} - C_{\text{leading}}}{\text{MARGIN}} = 1 - \frac{1}{1.040} \approx 3.85\%, \quad C_{\text{higher}} \text{ must be } < 0.040 C_{\text{leading}}.$$

A tail bounded only by $\sim 2 C_{\text{sunset}} = O(C_{\text{leading}})$ is roughly an order of magnitude too large to be admitted into a 3.85% slack. The screened-finiteness therefore does NOT establish that the higher skeletons respect the thin margin; it only establishes that they do not diverge.

5. Verdict and residual

ACCEPTED. The self-energy/free-energy correction (Section 2): the certificate quantity is the free-energy ratio, whose leading value is the SC-SCOPE thin joint $\times 1.040$ – $\times 1.13$.

RETRACTED. v1.0’s “RES-5 survives at STRONG EVIDENCE, thin”: the higher-skeleton tail is same-order, not sub-dominant.

RES-5 / GAP-2: OPEN. What remains:

- the rigorous normalization identity $|\Delta\Gamma_{2\text{-loop}}^{\text{pd}}|/\Delta F_{\text{margin}} = C_{\text{SC}}$ (layer-margin/cost bookkeeping $\leftrightarrow$ skeleton free-energy ratio);
- a quantitative higher-skeleton TAIL BUDGET small enough for the thin slack, $C_{\text{higher}} < \Delta F_{\text{margin}} - C_{\text{leading}}$, i.e. $C_{\text{higher}} \lesssim 0.04 C_{\text{leading}}$ – NOT the current same-order $C_{\text{higher}} \lesssim 2 C_{\text{sunset}}$.

No tier flip: B1-RH-ENUM stays T6 CONDITIONAL on {H-LAYER}. The 2PI common-mode mechanism and the screened ($1/(1+g)$, not $1/(1-g)$) response are real structural gains carried forward; they bound the tail finite but not yet thin.

6. Devil’s-advocate / self-adversarial review (CLAUDE.md 6.3)

- (α) “The $ca_0 \sim 0.002$ was wrong.” UPHELD and corrected (Section 2): it double-counted a_0 . Retracted as R-2026-06-09-res5-ca0-doublecount.
- (β) “The SC-SCOPE-joint $\rightarrow \Delta\Gamma_2^{\text{pd}}$ identity is asserted, not proved.” VALID: carried as the first residual (Section 3).
- (γ) “The screened tail $\leq 2 \times$ leading does not fit a 3.85% slack, so RES-5 does not survive.” UPHELD (the binding objection): the tail is same-order; v1.0’s survival verdict is retracted; RES-5/GAP-2 is OPEN; the next note must produce a quantitative tail budget $C_{\text{higher}} \lesssim 0.04 C_{\text{leading}}$. Registered as AUDIT-2026-06-09-res5-survival-overclaim.

Quantitative sanity check (mandatory): *slack* $= 1 - 1/1.040 = 0.0385$; *current tail bound* $\approx 2 C_{\text{sunset}} = O(C_{\text{leading}})$; *ratio* (tail bound)/(slack budget) $\approx 2 C_{\text{sunset}}/(0.04 C_{\text{leading}}) \gg 1$ – the tail is NOT within the slack, so RES-5 is OPEN (consistent with the retraction, not with v1.0’s survival).

7. Result footer

Result ID: B1-RH-ENUM / RES-5 sunset-norm certificate v1.1 (audit re-issue)

Precise statement: ACCEPTED: the a0-skeleton c a0 ~ 0.002 double-counted a0; the certificate quantity is $|\Delta \Gamma_2^{\text{pd}}|/\Delta F_{\text{margin}}$, leading value = SC-SCOPE joint x1.040->x1.13. RETRACTED: v1.0's "RES-5 survives STRONG EVIDENCE thin". The screened higher-skeleton tail $\leq 2x$ leading is SAME-ORDER, not sub-dominant; against the thin joint x1.040 the slack is $\sim 3.85\%$ (C_{higher} must be $< 0.04 C_{\text{leading}}$), which a same-order tail does not respect. RES-5/GAP-2 = OPEN.

Scope: Correction + honest downgrade. The normalization identity and a thin-slack tail budget are OPEN.

Dependencies: res5-a0-skeleton-sensitivity (screened response); SC-SCOPE (joint x1.040-x1.13); supersedes v1.0.

Evidence grade: ANALYTIC (correction + slack arithmetic) + EXECUTED (res5_sunset_norm_map.py).

Reproduction command: python codes/vacuum/res5_sunset_norm_map.py

Expected output: joint x1.040-x1.13>1; tail x1.97 (same-order); slack 0.0385; tail NOT within slack -> RES-5 OPEN; asserts PASS.

Falsification gate: A proved tail budget $C_{\text{higher}} < 0.04 C_{\text{leading}}$ (would re-open the survival route); or a normalization identity giving the leading ratio ≥ 1 (selection overturned).

Tier before / after: No tier flip. B1 T6 on {H-LAYER}. RES-5/GAP-2 OPEN (downgraded from v1.0's STRONG EVIDENCE thin by operator adversarial review 2026-06-09).

No-overclaim statement: This note CORRECTS a self-energy/free-energy conflation (accepted) and RETRACTS the survival verdict (RES-5 OPEN). It does NOT close RES-5 and does NOT flip B1. The 2PI common-mode + screened response are carried forward as finite-but-not-thin tail control.

Next required action: res5-tail-budget-closure: prove the normalization identity and a tail budget $C_{\text{higher}} < \Delta F_{\text{margin}} - C_{\text{leading}}$ ($\sim 0.04 C_{\text{leading}}$); only then can RES-5 -> controlled-error at the thin SC-SCOPE grade.